A 46 μW 13 b 6.4 MS/s SAR ADC with Background Mismatch and Offset Calibration

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A 46 μW 13 b 6.4 MS/s SAR ADC With Background Mismatch and Offset Calibration

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***Abstract*—A 6.4 MS/s 13 b ADC with a low-power background calibration for DAC mismatch and comparator offset errors is presented. Redundancy deals with DAC settling and facilitates calibration. A two-mode comparator and 0.3 fF capacitors reduce power and area. The background calibration can directly detect the sign of the dynamic comparator offset error and the DAC mismatch errors and correct both of them simultaneously in a stepwise feedback loop. The calibration achieves 20 dB spur reduction with little area and power overhead. The chip is implemented in 40 nm CMOS and consumes 46 μW from a 1 V supply, and achieves 64.1 dB SNDR and a FoM of 5.5 fJ/conversion-step at Nyquist.**

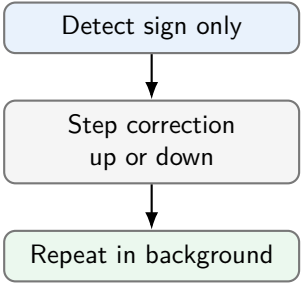
Introduction

- Wireless standards such as **IEEE 802.15.4g** need **high-resolution** ADCs at MS/s rates with **very low power**.
- SAR ADCs are attractive because they are already very **energy efficient**.
- Above roughly **10 bits**, the main limits become **DAC mismatch** and **noise / comparator offset**.
- Larger capacitors improve matching, but hurt **power**, **area**, and **speed**.
- Prior fixes such as **DSP/LMS**, **statistics-based methods**, pipelined SAR + amplifier, or comparator oversampling all add overhead.
- This paper instead proposes a **13-bit SAR ADC** with a **reconfigurable comparator** and **low-power background calibration** for DAC mismatch and dynamic comparator offset.

Reported results	
Resolution	13 b
Sampling rate	6.4 MS/s
Power	46 μW
ENOB	10.4 b
SNDR	64.1 dB
FoM	5.5 fJ/conv-step

Design Philosophy

- **Minimalist calibration**: do the least computation needed for the maximum return.
- Do **not** estimate full error magnitude with heavy DSP.
- Detect only the **sign / direction** of the error.
- Let a lightweight controller repeatedly nudge the hardware in the right direction.
- Result: an elegant, low-overhead **closed-loop control system** inside the SAR ADC.



Architecture and Closed-Loop Idea

Normal SAR conversion → Detect special codes → Compare two equivalent states → Detect sign of error → Adjust hardware and repeat

- Standard differential SAR ADC with **monotonic switching**.
- Uses **15 comparison cycles** for a **13-bit** conversion.
- Two **redundant cycles** make calibration observable and tolerate DAC settling / comparator error.
- An optional **16th cycle** is inserted only for calibration.
- Final output is computed from the **15-bit raw code**.

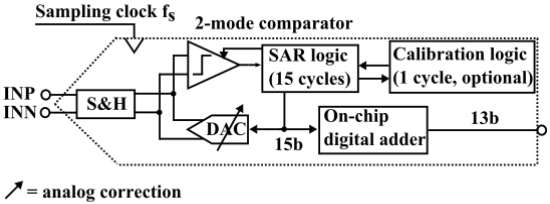


Fig. 2. Block diagram of a 13 b SAR ADC.

Digital Side: Lightweight Controller

1. **Sense & Force trigger logic**
   - Monitor specific internal SAR codes during the normal **15-cycle** conversion.
   - Implemented with **dynamic logic**; effectively "asleep" about **90%** of the time.
2. **Extra 16th cycle**
   - If a trigger code appears, the controller inserts an extra **16th cycle**.
   - **DAC mismatch**: compare two redundant codes A and B that should produce the same voltage.
   - **Comparator offset**: compare fine mode and coarse mode with the DAC code held constant.
3. **Sign detection and filtering**
   - Compare D_{15} and D_{16} to get only the **sign** of the error.
   - Feed that sign to a **6-bit bidirectional-counter** LPF.
   - Only when a counter reaches **0 or 63** does the analog calibration register update.

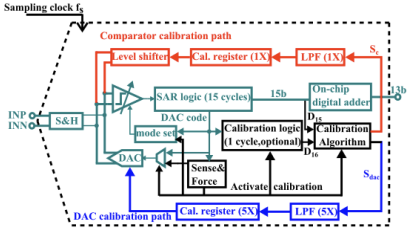


Fig. 10. Block diagram of the implemented 13 b SAR ADC.

Analog Side: Physical Error Correction

DAC mismatch correction

- Only the **top 5 MSB capacitors** are calibrated.
- Each has a programmable bank of tiny **75 aF** trim capacitors.
- These implement **0.25 LSB** correction steps by adding or subtracting capacitance from the main DAC bit weight.

Comparator offset correction

- A **2-mode comparator** saves power but creates a dynamic offset shift.
- Two programmable capacitor banks, C_a and C_b, inject a small compensating input step.
- Digital logic only chooses the control word; the actual correction happens **in analog**.

Key point: the digital side decides *when* and *which direction*; the analog side performs the actual fix.

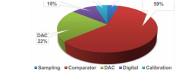


Fig. 17. Power consumption breakdown of the chip.

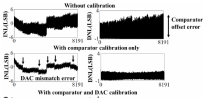


Fig. 19. Measured spectrum of the chip in three scenarios: without calibration, with comparator offset calibration, and with both calibrations.

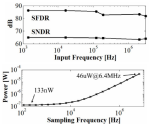


Fig. 20. Measured dynamic performance of the chip.

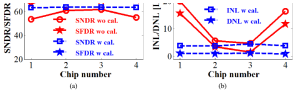
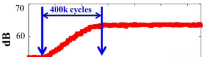


Fig. 21. Measured performance of four samples: (a) SNDR/SFDR and (b) INL/DNL.



detect the sign of the errors and apply feedback to the analog correction circuits, correcting both the comparator dynamic offset error and the DAC mismatch error at the same time. Thanks to the simple calibration algorithm and the custom-designed dynamic logic, the ADC achieves 5.5 Ωconv.step at 6.4 MS/s with negligible overhead in area and power.

Measurements Summary

Measured / Reported Results	
Process	40 nm CMOS
Supply voltage	1 V
Active area	0.0675 mm^2
Digital calibration area	0.0017 mm^2
Analog correction area	0.0009 mm^2
Sampling rate	6.4 MS/s
Resolution	13 b
Total power	46 μW
Calibration power overhead	$\sim$5%
ENOB	10.4 b
SNDR	64.1 dB
SFDR	81.9 dB
FoM	5.5 fJ/conv-step
Spur reduction from calibration	$\sim$20 dB
Convergence time	$\sim$400k cycles